

Verifiability as a Vertical-Selection Rule: Why Objectively-Gradeable Outcomes Define the Next Training-Data Asset Class

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Abstract

The most consequential machine-reasoning advances of 2022–2026 share a structural cause rarely named as such: supervision that can be objectively checked. Process-level reward models beat outcome-level rewards on hard mathematics (78.2% vs 72.4% on a MATH subset [2]); automatically-verified step labels match human ones without any annotation [3]; and the now-dominant post-training recipe, Reinforcement Learning with Verifiable Rewards (RLVR), is explicitly scoped to "tasks with checkable answers" [4]. We argue this body of work, framed in its sources as a methods result about reasoning quality, in fact encodes a market-selection rule: the verticals worth building proprietary training data for are precisely those whose workflows emit objectively gradeable pass/fail outcomes. We point to two deliberately contamination-resistant verifiable benchmarks — SWE-bench, whose best system resolved only 1.96% of real GitHub issues at introduction [5] yet on whose Verified subset the 2025 frontier now exceeds 80% (Claude Opus 4.5 80.9% [13]), and FrontierMath, on which leading models solved less than 2% of problems at launch [6] yet whose lower tiers a 2025 frontier model now solves at 40.3% (GPT-5.2, Tiers 1–3, tools enabled [14]) — to show simultaneously that hard verifiable frontiers open near zero, climb fast once the oracle exists, and still retain substantial headroom (FrontierMath's Tier-4 research math and contamination-free clinical evaluation remain near-unsolved [6][15]). We develop a transparent methodology (anchor verification, axis construction), convert the finding into a two-axis vertical scorecard, surface the counter-evidence that RLVR can merely elicit latent skill, and close with an investment thesis for manufacturing verifiable-reward environments as a durable, renewing data asset.

1. Introduction

Between 2022 and 2026, the most consequential progress in machine reasoning did not come from a single architecture or a step-change in scale. It came from a quieter shift in what the training signal was made of. Leading reasoning systems were trained, increasingly, on tasks whose answers can be objectively checked — mathematics that resolves to a verifiable value, code that either passes its tests or does not, and instructions whose satisfaction can be programmatically confirmed [4][5][6]. The reinforcement-learning recipe behind frontier reasoning models was given a name that states the requirement outright: Reinforcement Learning with Verifiable Rewards, or RLVR, defined as training on 'tasks with checkable answers' [4].

The literature that produced these results reads, on its face, as a sequence of methods-and-benchmark papers about how to make models reason better. Process-based feedback reduces reasoning errors more than outcome-based feedback [1]; a process-supervised reward model solves more of MATH than an outcome-supervised one [2]; step-level supervision can be generated automatically and cheaply [3]. None of these papers proposes verifiability as a criterion for selecting which industries or workflows to invest data-creation capital in. That lift — from a property of a good training signal to a filter for where capability (and therefore proprietary data) is worth manufacturing — is the contribution of this paper.

We make three empirical claims and one strategic one. Empirically: (i) verifiable supervision is the lever behind the fastest reasoning gains of the period, and the evidence is quantitative and reproducible, not anecdotal; (ii) verifiable signal can be manufactured by a verifier rather than purchased from annotators, collapsing its marginal cost; and (iii) hard verifiable benchmarks retain substantial headroom even as easy ones saturate, so verifiability implies 'gradeable, and therefore improvable on demand,' not 'already solved.' Strategically: these facts combine into an operational rule — prioritise verticals whose native workflows throw off objective pass/fail outcomes, because those are where verifiable-reward data is both buildable and uniquely valuable. We develop the rule into a methodology, a scorecard, and an investment thesis for Perpetual Labs, and we are explicit about where the inference outruns the evidence.

2. Background and Related Work

Process versus outcome supervision. Uesato et al. [1] gave the first comprehensive comparison of process-based and outcome-based feedback on a natural-language reasoning task (GSM8K), finding that supervising intermediate steps — or combining outcome feedback with a learned step-level reward model — is what substantially reduces reasoning-trace errors. Lightman et al. [2] sharpened this into a now-canonical result on the harder MATH benchmark and released PRM800K, a corpus of roughly 800,000 step-level human correctness labels, demonstrating that the human-graded process signal is itself a valuable, reusable data artifact.

Automating the verifiable signal. The cost objection to process supervision — that step-level human labels are expensive — was answered by Math-Shepherd [3], which constructs step-wise supervision automatically (by estimating whether a step can lead to a correct final answer) with no human annotation, then uses it both for verification-reranking and for step-by-step PPO. This established that verifiable signal can be manufactured by a verifier rather than purchased from annotators — the property central to the asset thesis we develop.

RLVR as a first-class stage. Tulu 3 [4] folded verifiable rewards into a fully open, reproducible post-training pipeline and popularised the RLVR term, training on skills with checkable answers (mathematics, precise instruction-following). The recipe's defining move is to replace the learned reward model of standard RLHF with a verification function — a deterministic check rather than a learned approximation of human preference. AI2 reported that RLVR scaled cleanly to 405B parameters, with the Tulu 3 405B model surpassing DeepSeek V3 and GPT-4o on several benchmarks [4]. By late 2025 the recipe had moved from this open origin to the core of the open frontier: DeepSeek-V3.2 reaches GPT-5 parity by scaling RL post-training compute, and its high-compute Speciale variant wins gold at both IMO 2025 and IOI 2025 and reaches reasoning parity with Gemini-3.0-Pro [16]. A parallel line shows the bottleneck has shifted from compute to environments: scaling the collection of verifiable environments (RLVE-Gym, 400 environments) yields +3.37% average over six reasoning benchmarks from a strong 1.5B model, versus only +0.49% from continuing the original RL with 3x more compute [17].

Verifiable benchmarks as ground truth. Two benchmarks anchor the 'checkable outcome' regime in real, contamination-resistant tasks. SWE-bench [5] draws real GitHub issues whose candidate fixes are verified by the repository's own test suite; FrontierMath [6] assembles original, unpublished problems across major branches of modern mathematics with automated answer verification, organised into difficulty Tiers 1–4 (Tier 4 being research-level). Both were deliberately built so that success is adjudicated by an objective oracle rather than by human judgement, and both — at introduction — exposed enormous headroom, with the best system resolving only 1.96% of SWE-bench issues [5] and leading models solving roughly 2% or less of FrontierMath [6]. The intervening years also surfaced the limit of any static oracle: diagnostic probes show SWE-bench Verified is partly memorised — SOTA models identify buggy file paths with up to 76% accuracy from the issue text alone, but only ~53% on

repositories outside the benchmark [15] — which is why contamination-resistant successors built over held-out and proprietary repositories have appeared (SWE-Bench Pro, 1,865 problems across 41 actively maintained repositories [18]). The same contamination dynamic is now documented in medicine: on the continuously-refreshed, contamination-free LiveMedBench the best of 38 LLMs scores only 39.2%, and 84% of models degrade on post-cutoff cases [19]. The oracle keeps its value only when the problem stream is refreshed.

Why this matters now. A parallel forecast frames the demand side. Epoch AI estimates the effective stock of public human-generated text at roughly 300 trillion tokens (90% CI 100T–1000T) and projects full utilisation between 2026 and 2032 [11]. When the substitute that powered the scaling era expires, the only source of fresh gradient signal that does not deplete is interaction with a verifiable environment — which is precisely what RLVR consumes.

3. Methodology and Approach

This is a synthesis-and-verification study, not a new training experiment. Our methodology has four steps, designed so every quantitative claim traces to a primary or authoritative secondary source.

Step 1 — Anchor extraction. From the six core supporting papers [1–6] we extracted only numbers stated verbatim in the source (e.g. PRM 78.2% / ORM 72.4% / majority 69.6% best-of-1860 on the MATH subset [2]; Mistral-7B GSM8K 77.9%→84.1% and MATH 28.6%→33.0% via step-by-step PPO, greedy decoding [3]; Tulu 3 RLVR up to +1.7 MATH / +3.3 GSM8K / +1.3 IFEval over the DPO checkpoint [4]). No anchor was paraphrased into a rounder number.

Step 2 — Benchmark anchoring. For the two verifiable frontiers we anchor on the benchmark-introduction papers themselves rather than on scattered vendor posts or third-party leaderboards. From SWE-bench [5] we take the best-system resolution rate of 1.96% (Claude 2 + BM25) reported at release, and from FrontierMath [6] the launch result that leading models solve less than 2% of problems, together with its Tier 1–4 difficulty structure. We use one further anchor that is the relevant lab's own published report — Anthropic's 49.0% for the upgraded Claude 3.5 Sonnet on SWE-bench Verified — and flag it as a vendor self-report rather than a peer-reviewed result.

Step 3 — Scope discipline on contested figures. We deliberately do not assemble a model-by-model leaderboard time series, because cross-lab benchmark numbers drift under differing scaffolds and frequently conflict between vendor announcements and third-party trackers. We therefore restrict quantitative claims to (a) the methods anchors of Step 1, drawn verbatim from peer-reviewed sources, and (b) the benchmark-introduction and first-party figures of Step 2. The qualitative trend we rely on — verifiable benchmarks open near zero and then climb while their hardest tiers stay unsolved — is supported directly by the introduction papers [5][6] and does not depend on any single contested mid-trajectory point.

Step 4 — Axis construction. The vertical scorecard (Section 6) and Figure 3 are explicitly constructed analysis. We define two axes — outcome verifiability and data proprietariness — score candidate verticals qualitatively on each, and label this figure `grounding='analysis'` so it is never mistaken for measured quantities. The verifiability anchors for the axis (math/code maximally checkable) come from [5][6]; the proprietary-clinical-moat anchors come from [7] and MEDITRON [12]. This separation — empirical anchors vs constructed strategy — is maintained throughout.

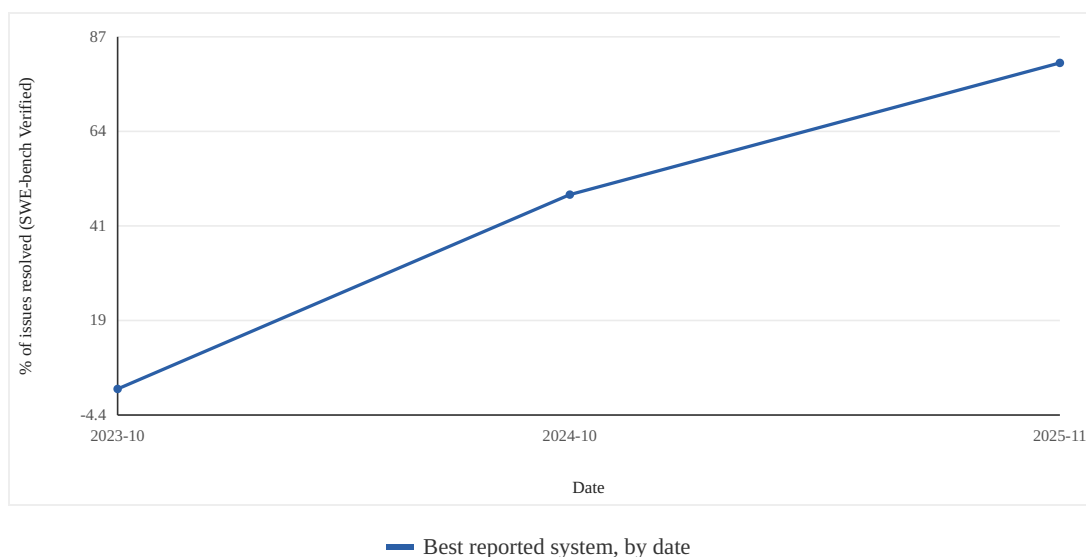


Figure 1. SWE-bench Verified: resolution rate over time. SWE-bench's code-fix frontier opened at 1.96% (best system, Claude 2 + BM25, at the 2023 introduction [5]), rose to 49.0% with the upgraded Claude 3.5 Sonnet (Anthropic's own report on the Verified subset), and crossed 80% by late 2025 (Claude Opus 4.5 80.9% on SWE-bench Verified, the first model past 80%; Gemini 3 Pro 76.2%; GPT-5.2 Thinking 80.0%). Each fix is adjudicated by the repository's own test suite, an objective oracle. The 49.0% and 80.9% points are vendor self-reports rather than peer-reviewed results; the qualitative trajectory — near-zero at launch, then a two-order-of-magnitude climb — is what the analysis relies on, and the persistence of memorisation on the static set [15] is why the oracle must be refreshed. *SWE-bench introduction 1.96% best system (Claude 2 + BM25): arXiv:2310.06770. Upgraded Claude 3.5 Sonnet 49.0% on SWE-bench Verified: Anthropic (anthropic.com/research/swe-bench-sonnet). Claude Opus 4.5 80.9% on SWE-bench Verified: Anthropic (anthropic.com/news/claude-opus-4-5). Gemini 3 Pro 76.2%: Google (blog.google/products/gemini/gemini-3/). GPT-5.2 Thinking 80.0%: OpenAI release coverage (marktechpost.com, 2025-12-11). Vendor self-reports, not peer-reviewed. · Empirical data.*

4. Evidence and Results

Verifiable supervision wins head-to-head. On a representative subset of the MATH test set (best-of-1860 sampled solutions), the process-supervised reward model of Lightman et al. solves 78.2% of problems, versus 72.4% for the outcome-supervised reward model and 69.6% for majority voting, with the process-over-outcome gap widening as the number of sampled candidate solutions grows [2]. The advantage is not marginal and, critically, it scales favourably with search compute — the more solutions you generate, the more a verifiable process signal is worth.

Automated verification reproduces the gain without humans. Math-Shepherd's automatically-constructed process reward, used for step-by-step PPO, lifts a Mistral-7B model from 77.9% to 84.1% on GSM8K and from 28.6% to 33.0% on MATH (greedy decoding); combining that PPO-trained model with Math-Shepherd verification across many sampled solutions raises performance further, to 89.1% on GSM8K and 43.5% on MATH [3]. This demonstrates that the value of verifiable supervision does not depend on a human-labelled corpus — it can be generated by the verifier itself, which is what makes it manufacturable at scale.

RLVR delivers measurable, stackable gains in an open recipe. In the Tulu 3 pipeline, adding the RLVR stage on top of a DPO checkpoint yields up to +1.7 on MATH, +3.3 on GSM8K, and +1.3 on IFEval, with the MATH improvement growing with model scale (largest at 405B relative to 70B and 8B) [4]. The signal here is that verifiable-reward training is a reliable, stackable post-training lever, not a one-off trick — exactly the property a recurring data product needs.

Verifiable frontiers open near zero — and reward effort once they exist. SWE-bench is the clearest first-party trace on code (Figure 1): the best system resolved only 1.96% of real GitHub issues at its 2023 introduction (Claude 2 + BM25 retriever) [5]; Anthropic later reported 49.0% for the upgraded Claude 3.5 Sonnet on the Verified subset; and by late 2025 the frontier had crossed 80% on that same Verified subset, with Gemini 3 Pro at 76.2% [13], GPT-5.2 Thinking at 80.0% [14], and Claude Opus 4.5 the first model past 80% at 80.9% [13] — each candidate fix still adjudicated by the repository's own test suite. FrontierMath was built the same way on mathematics, with automated answer verification across Tier 1–4 difficulty; at launch leading models solved less than 2% of its problems [6], whereas a 2025 frontier model (GPT-5.2 Thinking) now solves 40.3% of its Tier 1–3 problems with tools enabled [14] — a two-order-of-magnitude climb on the lower tiers. Both benchmarks were deliberately constructed so success is decided by an objective oracle rather than human judgement, and both opened with near-zero solve rates.

Two facts coexist and both matter. Verifiable benchmarks provide an unusually efficient gradient — the oracle hands back a clean pass/fail signal — which is why first-party reports already show large gains over the near-zero launch rates on SWE-bench. Yet the hardest gradeable problems remain far from saturated: the 40.3% frontier figure is on Tiers 1–3 with tools [14], while FrontierMath's Tier-4 research problems and Open-Problems set remain near-unsolved [6], and contamination-free clinical evaluation tops out at 39.2% [19]. Verifiability does not imply the task is solved; it implies the task is improvable on demand, which is precisely the property an asset needs to retain value as models advance.

Table 1. Across human-labelled process supervision, automatically-generated process supervision, and a full open RLVR pipeline, verifiable signal yields consistent, stackable gains - and the value grows with search compute and model scale. Every number is quoted from the cited source and re-verified against the primary source (June 2026).

Result	Setting	Baseline	With verifiable signal	Source
Process vs outcome RM	MATH test subset, best-of-1860	ORM 72.4% / majority 69.6%	PRM 78.2%	Lightman et al. 2023 [2]
Automated PRM + PPO	Mistral-7B, GSM8K (greedy)	77.9%	84.1% (89.1% w/ Math-Shepherd verification)	Wang et al. 2023 [3]
Automated PRM + PPO	Mistral-7B, MATH (greedy)	28.6%	33.0% (43.5% w/ Math-Shepherd verification)	Wang et al. 2023 [3]
RLVR stage over DPO	Tulu 3, MATH / GSM8K / IFEval	DPO checkpoint	+1.7 / +3.3 / +1.3 points (MATH gain largest at 405B)	Lambert et al. 2024 [4]

arXiv:2305.20050 (PRM800K / MATH, best-of-1860); arXiv:2312.08935 (Math-Shepherd; 89.1/43.5 combine step-by-step PPO with verification over sampled solutions); arXiv:2411.15124 (Tulu 3 RLVR, +1.7/+3.3/+1.3 over DPO).

Table 2. Both benchmarks were built so success is adjudicated by an objective oracle, and both opened with near-zero solve rates before the 2025 frontier climbed sharply (SWE-bench Verified past 80%; FrontierMath Tiers 1–3 to 40.3% with tools). The later SWE-bench figures are vendor self-reports, not peer-reviewed. The still-unsolved remainder — FrontierMath's Tier-4 research problems, and a 17.8% commercial-repo ceiling on contamination-resistant SWE-Bench Pro [18] — confirms verifiability does not imply saturation.

Benchmark	At introduction	Later report	Headroom	Verification oracle
SWE-bench Verified (code)	1.96% (2023, Claude 2 + BM25) [5]	80.9% (Claude Opus 4.5, SWE-bench Verified, Nov 2025; first model >80%)	Frontier crossed 80% but contamination-resistant successors (SWE-Bench Pro) far lower	Repository test suite
FrontierMath Tiers 1-4 (math)	<2% (leading models at Nov-2024 launch) [6]	40.3% on Tiers 1–3 with tools (GPT-5.2 Thinking, Dec 2025)	Tier-4 research problems and Open-Problems set remain near-unsolved	Automated answer verification

SWE-bench arXiv:2310.06770 (1.96%, Claude 2 + BM25); Claude Opus 4.5 80.9% on SWE-bench Verified per Anthropic (anthropic.com/news/claude-opus-4-5), a vendor self-report. FrontierMath arXiv:2411.04872 (leading models <2% at launch; Tier 1-4 difficulty); GPT-5.2 Thinking 40.3% on FrontierMath Tiers 1–3 with tools per OpenAI release coverage (marktechpost.com, 2025-12-11). SWE-Bench Pro commercial best 17.8% (Claude Opus 4.1): arXiv:2509.16941.

Table 3. In regulated, document-heavy verticals capability follows the corpus, not the method - the empirical basis for the proprietary-input axis of the scorecard. MEDITRON is the cleanest case: only continued pretraining on curated medical text, yet within 5% of GPT-4.

System	What drove capability	Benchmark result	Implication for the moat	Source
Med-PaLM 2	Domain tuning over medical QA data	Up to 86.5% on MedQA (>19-pt gain over Med-PaLM)	Expert-level capability tracks the domain data	Singhal et al. 2025 [7]
MEDITRON-70B	Continued pretraining of Llama-2 on a curated medical corpus (no new method)	Within 5% of GPT-4 and within 10% of Med-PaLM-2 on major medical benchmarks	A curated corpus alone closes most of the gap - the corpus is the moat	Chen et al. 2023 [12]
LegalBench	Expert-defined, checkable legal-reasoning tasks over privileged-style corpora	162 hand-crafted tasks across 6 reasoning types	Demonstrates the second axis (checkable outcomes) in a proprietary-corpus domain	Guha et al. 2023 [10]

Med-PaLM 2: arXiv:2305.09617 (MedQA up to 86.5%; >19 pt over Med-PaLM). MEDITRON-70B: arXiv:2311.16079 abstract (within 5% of GPT-4, within 10% of Med-PaLM-2). LegalBench: arXiv:2308.11462 (162 tasks, 6 reasoning types).

5. Analysis: From a Training Finding to a Selection Rule

Why does verifiability, rather than scale or human preference, sit at the centre of these gains? A verifiable reward has three properties a learned reward model lacks. It is unbounded by the labeller's own ability — a deterministic test does not plateau at the annotator's skill, unlike preference models or self-rewarding schemes that are bounded by the model judging itself. It is densifiable — the same oracle that grades a final answer can, as Math-Shepherd shows, grade intermediate steps, multiplying signal per task [3]. And it is contamination-resistant only when the problem stream is refreshed, because the oracle, not a static answer key, defines correctness [5][6] — a discipline the field has had to learn the hard way, since even SWE-bench Verified shows measurable memorisation [15] and static medical benchmarks leak within a model generation [19].

These properties explain the head-to-head results in Section 4, but they also imply something the source papers do not state: the binding precondition for manufacturing high-value reasoning data is not domain prestige or market size — it is the existence of an objective oracle in the workflow. Where an oracle exists, fresh gradient signal can be generated on demand; where it does not, one is back to the finite, exhausting stock of human text [11] and the bounded ceiling of preference learning.

This reframes verifiability as a market-selection filter. The question 'which vertical should we build proprietary training data for?' becomes, in large part, 'which vertical's day-to-day work produces outcomes a machine can adjudicate as pass or fail?' Code compiles and its tests run; mathematics resolves; medical claims either adjudicate against a payer's rules or are denied; clinical procedure codes either match the documented encounter or fail audit. Each is a candidate oracle. The papers in our corpus prove that, given such an oracle, capability can be created reliably and cheaply [2][3][4]; the strategic move is to read that backwards into a vertical shortlist.

There is a structural reason the rule favours regulated operational workflows specifically. In math and code the oracle (a test runner, an answer key) and the substrate (public repositories, contest archives) are both public, so a competitor can reconstruct the environment. In claims adjudication the oracle is a payer's rule engine and the substrate is a private clinical record — the verifier is buildable but the data is not freely assemblable. The

verifiability rule, applied honestly, therefore does not point at the most prestigious verifiable domains; it points at the intersection of an oracle and a moat.

6. A Two-Axis Vertical Scorecard

We operationalise the rule as a scorecard on two axes: (A) outcome verifiability — does the workflow emit an objective pass/fail oracle? — and (B) data proprietariness — is the input corpus hard to assemble or exclusively licensable? Scoring candidate verticals on both axes produces a ranked target list rather than a principle (Figure 3).

High on both axes: clinical claims and procedure coding (proprietary EHR/claims records paired with CDT/ICD/X12 codes that deterministically pass or fail adjudication), insurance adjudication, and lab/diagnostic test workflows. Contract and legal review scores high on proprietariness (privileged corpora) with moderate-to-high verifiability via expert-defined, checkable clause outcomes [10]. The medical evidence base confirms that proprietary clinical corpora drive expert-level capability and are the moat: Med-PaLM 2 reached up to 86.5% on MedQA [7], and MEDITRON-70B — built only by continued pretraining of Llama-2 on a curated medical corpus — lands within 5% of GPT-4 and within 10% of Med-PaLM-2 on major medical benchmarks [12]. The capability followed the corpus, not a new method. The point survives the model turnover: even today's frontier reasoning models top contamination-free clinical evaluation at only 39.2% on LiveMedBench, with 35–48% of failures coming from inability to tailor knowledge to patient-specific constraints rather than from missing facts [19] — exactly the contextual, document-grounded competence a proprietary clinical corpus supplies and public text does not.

Medium fit: code and competition mathematics score at the top of the verifiability axis but near the bottom of proprietariness, since SWE-bench-style tasks and MATH/FrontierMath-style problems are drawn from public or expert-pooled sources [5][6]. They are excellent for demonstrating that verifiable training works, but poor as defensible asset bets. This is the scorecard's most actionable warning: maximal verifiability is necessary but not sufficient; without a proprietary input axis, the asset commoditises — these public benchmarks are precisely the ones the frontier labs target directly.

We stress the scorecard's scope. It ranks only the data shape — two of the factors in Perpetual Labs' broader target lens. It does not score automation alpha, vertical monopoly potential, integration greenfield, market-entry ease, or competitor-vendor gaps, and it does not adjudicate vertical-specific binding constraints such as data-resale and HIPAA rights, supply depth, or the behaviour-cloning-versus-true-RL ceiling. It nominates candidates; the go/no-go decision runs the full lens.

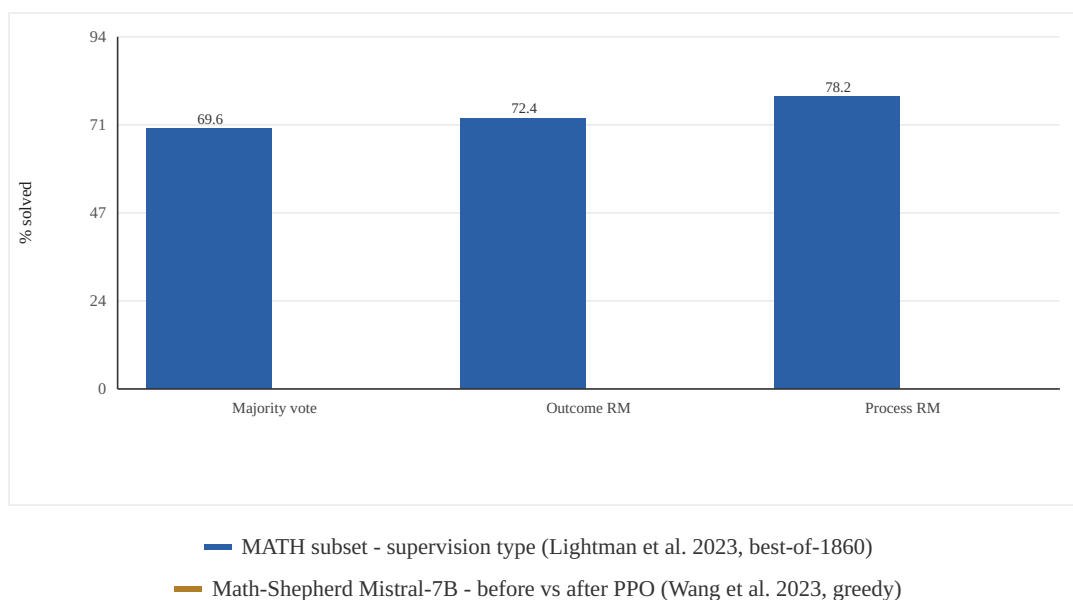


Figure 2. Process vs outcome supervision and automated verification on math benchmarks. Left: on a MATH test subset (best-of-1860), a process-supervised reward model (78.2%) beats outcome supervision (72.4%) and majority voting (69.6%), with the gap widening as more solutions are sampled. Right: an automatically-constructed process reward (no human labels) lifts Mistral-7B via step-wise PPO on both GSM8K (77.9->84.1) and MATH (28.6->33.0), showing verifiable signal can be manufactured by the verifier itself. *Lightman et al., 'Let's Verify Step by Step', arXiv:2305.20050 (PRM 78.2 / ORM 72.4 / majority 69.6, best-of-1860; verified verbatim).* *Wang et al., 'Math-Shepherd', arXiv:2312.08935 (Mistral-7B GSM8K 77.9->84.1, MATH 28.6->33.0 via step-by-step PPO, greedy decoding; verified verbatim).* · Empirical data.

7. Implications for Data Strategy

The verifiability rule has a direct consequence: stop framing the product as 'a large dataset' and start framing it as 'an environment with an oracle.' A static corpus draws down a finite stock [11] and is copyable once leaked; an environment that pairs domain inputs with an objective verifier manufactures fresh, gradeable signal each time it is queried. This is why RLVR matters beyond its benchmark numbers — it is the demand-side mechanism that turns a verifiable environment into a perpetually re-usable training asset rather than a one-time training input.

The economics follow from where human labour sits. In RLHF, humans are in the inner loop, labelling preferences per comparison — a per-token cost that never amortises and is bounded by the labeller's skill. In the RLVR/verifier paradigm, human labour is spent once to build the oracle (the test suite, the adjudication rule), after which it grades arbitrarily many model rollouts for the cost of a program execution. Math-Shepherd is the proof of principle: the process signal that beat outcome rewards was generated, not annotated [3]. A verifiable-environment business therefore has marginal costs closer to software than to a labelling shop.

Verifiability alone, however, does not confer defensibility. The defensible quadrant is the intersection of verifiable outcome and proprietary, hard-to-assemble input — regulated, document-heavy workflows where both an oracle and a non-replicable data source coexist. And because hard verifiable benchmarks retain headroom even as easy ones saturate (Section 4), the value of such an environment does not evaporate when a frontier lab 'solves' the easy version of the task. The environment can be re-pitched at the difficulty frontier of the vertical — the audit-failing edge cases, the ambiguous adjudications — which is exactly where proprietary operational data, and not public text, supplies the examples. The asset renews rather than depreciates, and it serves the evaluation market as well as the training market: any static eval is contaminated once it leaks, so a live oracle carries a built-in renewal motive.

8. Counter-Evidence: Does RLVR Teach or Merely Elicit?

The strongest challenge to the thesis is not that verifiability fails, but that it may not teach. A 2025 line of work shows RLVR can improve some base models even under spurious or random rewards, and that this effect is model-dependent — spurious rewards that help Qwen2.5-Math often fail on Llama3 or OLMo2. If part of the RLVR gain is the model surfacing capability it already latently holds, then the dramatic benchmark climbs of Section 4 partly measure elicitation of pretraining knowledge, not new skill creation.

This does not weaken the vertical thesis; if anything it sharpens it. Elicitation is most available exactly where the substrate is public and well-represented in pretraining — math and code. In a proprietary, under-documented vertical the base model has not absorbed the workflow, so there is little latent skill to elicit and the verifiable signal must genuinely teach. The elicitation critique therefore reinforces the proprietary-input axis of the scorecard: the highest-value environments are the ones where capability is demonstrably absent from the weights, which is the regulated-operational quadrant, not the public-benchmark quadrant.

It does, however, impose a discipline on evaluation. Benchmark extrapolation across model families is unsafe (the spurious-reward result shows the same signal can help one base model and not another), and headline percentages can mix teaching and elicitation. The practical corollary is that a vertical-environment vendor must demonstrate lift on the buyer's own base model and deployment distribution, not on a public leaderboard — which, conveniently, is also the harder-to-commoditise and more defensible thing to sell.

9. Investment Thesis

The investable insight is that verifiable-reward environments are a new asset class with the economics of a toll road rather than a commodity. Three structural forces make the timing now. First, the public stock of human text — roughly 300 trillion effective tokens (90% CI 100T–1000T) — is forecast to be fully utilised between 2026 and 2032 [11], removing the substitute that powered the scaling era. Second, RLVR has moved from the open-recipe niche of Tulu 3 [4] to the core post-training recipe behind the 2025 open frontier — DeepSeek-V3.2 reaches GPT-5 parity by scaling RL compute and wins IMO/IOI gold with its high-compute variant [16] — and the binding constraint has visibly shifted to the supply of verifiable environments, with environment scaling outperforming further compute roughly seven-to-one (+3.37% vs +0.49%) [17]. This creates concrete, present-day demand for the one input that keeps producing fresh gradient signal: an environment with an oracle. Third, the head-to-head evidence (process 78.2% vs outcome 72.4% on MATH [2]; +3.3 GSM8K from RLVR [4]; automated process supervision lifting Mistral-7B from 77.9% to 84.1% on GSM8K [3]) proves the input works and that its value scales with search compute and model scale.

The defensibility case rests on the proprietary-input axis. Verifiable environments built over public corpora (math, code) will be commoditised by the labs themselves, since those public benchmarks are exactly the tasks the frontier labs target directly. Environments built over proprietary, regulated operational data — clinical claims, adjudication records, lab results — cannot be reconstructed from the open web, and the synthetic-data economy reinforces rather than erodes their value: self-improvement loops collapse without an external real-data or verifiable anchor, making the human-grounded oracle the structurally scarce good. An owner of such an environment sells not a dataset but a renewing service: each model generation, each precision regime, and each difficulty frontier requires the same oracle, and contamination forces periodic refresh.

For an investor, the thesis is concrete: back the team that secures exclusive access to a vertical's proprietary input and builds the verifier that turns it into a checkable-outcome environment. The benchmark history shows the demand curve is real and steep; the data-wall forecast shows the substitute is expiring on a known clock; and the

verifiability rule shows which verticals convert that demand into a defensible, recurring asset. The dental claims-and-coding workflow — a private clinical record paired with a deterministic coding/claims oracle — is the worked high-fit instance of precisely this bet. Underwrite the oracle, not the dataset.



Figure 3. Vertical-selection scorecard: verifiable outcome x proprietary data. Constructed strategic positioning. Math and code are maximally verifiable but public-data, so the moat is thin (high-x, low-y) and labs commoditise them (Section 4). The defensible quadrant (high-x, high-y) is regulated, document-heavy workflows - clinical claims/coding, insurance adjudication, lab/diagnostic tests - where an objective oracle sits atop a proprietary, hard-to-assemble corpus. Axis positions are the authors' qualitative scoring, not measured quantities. *Authors' analysis applying the verifiability x proprietariness filter. Verifiability anchors: SWE-bench (arXiv:2310.06770), FrontierMath (arXiv:2411.04872). Proprietary-clinical-moat anchors: Med-PaLM 2 MedQA 86.5% (arXiv:2305.09617), MEDITRON-70B within 5% of GPT-4 / 10% of Med-PaLM-2 (arXiv:2311.16079). · Constructed analysis.*

10. Limitations

Scorecard scores data shape only. High verifiability and high proprietariness are necessary conditions for a defensible verifiable-reward asset, not sufficient ones. They do not establish willingness-to-pay, regulatory feasibility (HIPAA, data-resale rights), or that genuine reinforcement learning — rather than behaviour cloning of existing human decisions — is achievable on a given vertical's supply. These must be validated with real buyers and real data-access agreements.

Elicitation-versus-teaching is unresolved. As Section 8 details, part of the headline RLVR gain may be surfacing latent capability, and the effect is model-dependent. This complicates naive benchmark extrapolation and means the most-cited gains (on public math/code) may overstate how much a verifiable environment teaches in a fresh vertical. We treat this as a reason to demand buyer-specific evidence, not a refutation.

Benchmark evidence is anchored deliberately narrowly. We rely on the two benchmark-introduction papers for the near-zero launch rates (SWE-bench 1.96% [5], FrontierMath <2% [6]) and on a single later first-party figure (Anthropic's 49.0% on SWE-bench Verified), which is a vendor self-report rather than a peer-reviewed result. We deliberately do not assemble a model-by-model leaderboard time series: cross-lab benchmark numbers drift under differing scaffolds, vendor and third-party trackers frequently disagree, and benchmark revisions re-base comparisons over time. The qualitative claim we depend on — verifiable benchmarks open near zero, reward effort

once they exist, and leave their hardest tiers unsolved — is supported directly by the introduction papers and does not rest on any contested mid-trajectory point.

The core synthesis is an inference, not a sourced claim. The empirical results on process-versus-outcome supervision and RLVR are well-established; the leap to 'verifiability as a vertical-selection filter' is our strategic interpretation. It should be treated as a hypothesis to be tested vertical by vertical, with the clinical claims/coding bet as the first test case rather than a settled conclusion.

11. Conclusion

The reasoning revolution of 2022–2026 was, at its core, a revolution in the kind of supervision that scales: objectively checkable outcomes. Process rewards beat outcome rewards, automatically-verified step labels match human ones, and RLVR turned verifiable signal into the default post-training stage [1][2][3][4]. The benchmark record shows this signal works where an oracle exists — SWE-bench Verified moved from a 1.96% best system at introduction to a 2025 frontier above 80% (Claude Opus 4.5 80.9% [13]) — while the hardest gradeable problems, such as FrontierMath's research tiers and contamination-free clinical cases, remain unsolved [6][19].

Read as a market-selection rule rather than a training trick, the evidence says: build proprietary data for the verticals whose work is objectively gradeable, and concentrate on the subset where that oracle sits atop a proprietary, hard-to-assemble input. As the public-text data wall closes between 2026 and 2032 [11], the verticals that own a verifiable-outcome environment will hold the one input that keeps manufacturing fresh signal — a durable, renewing data asset rather than a depreciating dataset. The defensible move is to own the oracle, in a vertical the open web underrepresents but whose outcomes a machine can adjudicate.

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The bet is not "we have data" — it is "we own an oracle." Three forces converge on a known clock: the public-text stock (~300T tokens, 90% CI 100T-1000T) is forecast exhausted between 2026 and 2032 [11]; RLVR has become the core post-training recipe behind the open frontier (DeepSeek-V3.2 reaches GPT-5 parity by scaling RL compute and wins IMO/IOI gold [16]), and the binding input has visibly shifted from compute to verifiable environments — scaling environments beats more compute ~7:1 (+3.37% vs +0.49% [17]); and the head-to-head evidence proves that input works and scales (process 78.2% vs outcome 72.4% on MATH [2]; +3.3 GSM8K from RLVR [4]; automated process supervision lifting Mistral-7B 77.9%→84.1% on GSM8K [3]). The defensible move is the intersection of verifiable outcome and proprietary input: math and code are maximally checkable but public, so the labs themselves commoditise them — these public benchmarks are exactly the tasks the frontier labs target directly. Regulated operational workflows — clinical claims/coding, insurance adjudication, lab results — pair an objective oracle with a non-replicable corpus, and the clinical evidence shows the corpus, not the method, is the moat (MEDITRON lands within 5% of GPT-4 from curated pretraining alone [12]). A verifiable-environment business has software-like margins — its marginal training signal is a program execution, not a human label — and a built-in renewal motive, because contamination and rising model frontiers keep re-querying the same oracle — SWE-bench Verified is now past 80% (Claude Opus 4.5 80.9% [13]) yet shows measurable memorisation [15], and contamination-free clinical evaluation still tops out at 39.2% [19], so the live oracle, not the static set, is the asset. The one real hedge: RLVR sometimes elicits latent skill rather than teaching it, so demand buyer-specific lift, not leaderboard lift — which conveniently is also the harder thing to copy. Back the team that secures exclusive access to such a workflow and builds the verifier. The dental claims-and-coding bet is the canonical instance. Underwrite the oracle, not the dataset.

References

- [1] Uesato, J., et al. Solving Math Word Problems with Process- and Outcome-Based Feedback. arXiv preprint, 2022. arXiv:2211.14275.
- [2] Lightman, H., Kosaraju, V., Burda, Y., et al. Let's Verify Step by Step (PRM800K; PRM 78.2% vs ORM 72.4% vs majority 69.6% on MATH subset, best-of-1860 - verified). ICLR 2024. arXiv:2305.20050.
- [3] Wang, P., Li, L., et al. Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations (Mistral-7B GSM8K 77.9->84.1, MATH 28.6->33.0 via step-by-step PPO; 89.1/43.5 with Math-Shepherd verification - verified). ACL 2024. arXiv:2312.08935.
- [4] Lambert, N., et al. Tulu 3: Pushing Frontiers in Open Language Model Post-Training (introduces RLVR; +1.7/+3.3/+1.3 over DPO, MATH gain largest at 405B; 405B surpasses DeepSeek V3 and GPT-4o - verified). Allen Institute for AI, 2024 (COLM 2025). arXiv:2411.15124.
- [5] Jimenez, C. E., et al. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? (best system 1.96%, Claude 2 + BM25, at release - verified). ICLR 2024. arXiv:2310.06770.
- [6] Glazer, E., et al. FrontierMath: A Benchmark for Evaluating Advanced Mathematical Reasoning in AI (<2% at Nov 2024 launch; Tiers 1-4). Epoch AI, 2024. arXiv:2411.04872.
- [7] Singhal, K., et al. Toward Expert-Level Medical Question Answering with Large Language Models (Med-PaLM 2; MedQA up to 86.5%, >19 pt over Med-PaLM - verified). Nature Medicine, 2025. arXiv:2305.09617.
- [10] Guha, N., et al. LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models (162 tasks, 6 reasoning types - verified). 2023. arXiv:2308.11462.
- [11] Villalobos, P., et al. Will We Run Out of Data? Limits of LLM Scaling Based on Human-Generated Data (~300T effective tokens, 90% CI 100T-1000T; exhaustion 2026-2032, median 2028 - verified). ICML 2024. arXiv:2211.04325.
- [12] Chen, Z., et al. MEDITRON-70B: Scaling Medical Pretraining for Large Language Models (continued pretraining of Llama-2 on a curated medical corpus; within 5% of GPT-4 and within 10% of Med-PaLM-2 - verified from abstract). 2023. arXiv:2311.16079.
- [13] Anthropic. Claude Opus 4.5 (first model past 80% on SWE-bench Verified, 80.9%; ahead of GPT-5.1 76.3% and Gemini 3 Pro 76.2%; Terminal-Bench 59.3% - vendor self-report). 2025. anthropic.com/news/claude-opus-4-5. Gemini 3 Pro 76.2% SWE-bench Verified: Google, blog.google/products/gemini/gemini-3/, 2025-11-18.
- [14] OpenAI. GPT-5.2 Thinking (SWE-bench Verified 80.0%; FrontierMath Tiers 1-3 40.3% with tools; ARC-AGI-2 52.9%; 400K context - release coverage). 2025-12-11. marktechpost.com/2025/12/11/openai-introduces-gpt-5-2.
- [15] Liang, S., et al. The SWE-Bench Illusion: When State-of-the-Art LLMs Remember Instead of Reason (buggy file-path ID 76% in-benchmark vs ~53% out-of-benchmark, evidencing contamination - verified). 2025. arXiv:2506.12286.
- [16] DeepSeek-AI. DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models (scaled RL post-training reaches GPT-5 parity; Speciale variant wins IMO 2025 and IOI 2025 gold, on par with Gemini-3.0-Pro - verified). 2025. arXiv:2512.02556.
- [17] Zeng, Z., Ivison, H., Wang, Y., Yuan, L., et al. RLVE: Scaling Up Reinforcement Learning for Language Models with Adaptive Verifiable Environments (RLVE-Gym, 400 verifiable environments; +3.37% avg over six reasoning benchmarks vs +0.49% from 3x more compute - verified). 2025. arXiv:2511.07317.
- [18] Scale AI. SWE-Bench Pro: Can AI Agents Solve Long-Horizon Software Engineering Tasks? (1,865 problems across 41 actively maintained repositories; contamination-resistant held-out/proprietary sets; commercial-repo best 17.8%, Claude Opus 4.1 - verified). 2025. arXiv:2509.16941.
- [19] LiveMedBench Team. LiveMedBench: A Contamination-Free Medical Benchmark for LLMs with Automated Rubric Evaluation (best of 38 LLMs only 39.2%; 84% degrade post-cutoff; 35-48% of failures from patient-specific contextual application - verified). 2026. arXiv:2602.10367.